

Arthur Cancellieri Pires

Data Scientist / PhD Candidate

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Public project: Machine Anomaly Detection + Alarms

Professional Experience

- Railway Laboratory (Lafer) – partnership with VALE** On-site
TECHNICAL LEAD IN APPLIED AI *January 2022 - Present*
 - Predictive diagnosis of track defects:** Developed, using an unsupervised approach, Transformer-based autoencoders and a repeatability metric to detect and prioritize track defects in multivariate time series, validating the method on historical data with limited labels. The solution identified defects confirmed in the field, revealed cases absent from the maintenance database, and flagged issues weeks in advance of recorded interventions.
 - Monitoring of sensor failures and data quality:** Developed deep learning models and automated checks to detect failures in the instrumentation of EFVM wagons, achieving 91% precision on internally labeled events and reducing detection latency from 3 months to real time, enabling maintenance teams to target the component with the highest probability of failure.
 - Technical leadership of AI solutions:** Acted as technical lead for VALE's predictive maintenance program based on data from instrumented wagons, translating stakeholder demands into technical deliveries and coordinating a team of 12 members, including students and maintenance engineers.

Skills: Python + libraries (PyTorch, PyTorch Lightning, scikit-learn) · Deep Learning (Transformers, Autoencoders) · Anomaly Detection · Time Series · Deployment (FastAPI, Docker)
- Railway Laboratory (Lafer) – partnership with VALE** On-site
RESEARCH DATA SCIENTIST *June 2020 - January 2022*
 - Data pipeline for predictive models:** Designed and developed a pipeline for the daily ingestion and processing of raw data collected by instrumented wagons, generating between 20,000 and 40,000 records per wagon per day for machine learning model training and validation, while also integrating external VALE data sources to build an analytical dataset for predictive maintenance.
 - Track geometry estimation between inspections:** Developed deep learning models to estimate vertical track geometry from data collected by instrumented wagons, reducing the interval between specialized track geometry measurements. The solution achieved $R^2 \approx 0,98$ on the test set and supported more frequent track evaluation, reducing gaps between inspections.

Skills: Python + libraries (scikit-learn, PyTorch) · SQL, PostgreSQL · ETL · Deep Learning · Signal Processing
- Tribology and Railway Dynamics Laboratory (LabTDF) – partnership with VALE** On-site
INTERN IN MACHINE LEARNING AND OPTIMIZATION *Jan 2019 - June 2020*
 - Predictive modeling of the L/V ratio:** Developed a machine learning pipeline to estimate the wheel-rail contact L/V force ratio from onboard sensor time series, using multibody dynamics simulation and virtual sensors positioned at the same locations as the real sensors. The solution achieved $R^2 \approx 0,97$ and contributed to making VALE's predictive maintenance project with instrumented wagons viable.
 - Wheel-rail profile optimization for wear reduction:** Developed a wheel-rail profile optimization methodology using genetic algorithms and field data, resulting in a profile that reduced wear by 20%, maintained fatigue performance, and improved the L/V ratio by approximately 35%.
 - Operational recommendation for maintenance threshold:** Recommended revising EFVM's maintenance threshold from 3 mm to 2 mm based on the optimization results, projecting a service life increase of at least 29% for the current profile and 50% for the proposed profile. The recommendation was adopted by EFVM.

Skills: Python, MATLAB · Machine Learning (scikit-learn, Optuna) · Genetic Algorithms / Optimization · Multibody Dynamics (Universal Mechanism, Simpack)

Education

- University of Campinas (UNICAMP)** Campinas, Brazil
PH.D. IN MECHANICAL ENGINEERING *Expected: August 2026*
 - Thesis:** A methodology for detecting track defects from instrumented railway vehicle data.
- University of Campinas (UNICAMP)** Campinas, Brazil
M.SC. IN MECHANICAL ENGINEERING *Jan 2022*
 - Thesis:** Measuring railway track irregularities from instrumented railway vehicle data using machine learning techniques.
- Federal University of Espírito Santo (UFES)** Vitória, Brazil
B.SC. IN MECHANICAL ENGINEERING *May 2020*
 - Thesis:** Indirect identification of wheel-rail contact forces of an instrumented heavy haul railway vehicle using machine learning.

Languages

- Portuguese, Native
- English, Fluent
- Italian, Basic

Publications & Conferences

PUBLICATIONS

- 2024 **Measuring vertical track irregularities from instrumented heavy haul railway vehicle data using machine learning**, Engineering Applications of Artificial Intelligence
- 2023 **Influence of wheel tread wear on Rolling Contact Fatigue and on the dynamics of railway vehicles**, Wear
- 2022 **Nonintrusive Operator Inference for Chaotic Systems**, IEEE Transactions on Artificial Intelligence
- 2021 **Indirect identification of wheel–rail contact forces of an instrumented heavy haul railway vehicle using machine learning**, Mechanical Systems and Signal Processing
- 2021 **The effect of railway wheel wear on reprofiling and service life**, Wear

CONFERENCE PRESENTATIONS

- 2025 International Heavy Haul Association (IHHA) Conference, Nov. 17–21 *Colorado Springs, USA*
- 2023 International Heavy Haul Association (IHHA) Conference, Aug. 27–31 *Rio de Janeiro, Brazil*
- 2023 Railway Engineering Symposium, May 19–20 *Campinas, Brazil*